Linking Foundational Exoplanetary Physics to Habitability Inference

NASA HWO Habitability Explorer Project

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Abstract

We examine physically derived exoplanet features—including equilibrium temperature, irradiation scaling, escape velocity ratio, tidal locking propensity, and composite similarity indices—to elucidate their statistical structure and explanatory power for a learned habitability score. Using distributional diagnostics, correlation structure, response surface inspection, binned trend aggregation, discriminative margin analysis, and model-based feature importance, we identify principal drivers, redundancies, and limiting degeneracies. Results highlight radiative balance and atmospheric retention proxies as dominant determinants, while multicollinearity among bulk parameters suggests scope for dimensionality reduction.

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1 Introduction

The characterization of exoplanet habitability necessitates translating catalog observables (stellar effective temperature, stellar radius, planetary radius, mass proxies, orbital semi-major axis) into physics-grounded descriptors. These descriptors (e.g., equilibrium temperature T_{eq}, escape velocity v_{esc}, insolation flux S, tidal locking likelihood, and the Earth Similarity Index (ESI)) capture energy balance, retention capacity, and dynamical state. This manuscript formalizes previously exploratory interpretation, supplying academically styled exposition suitable for archival review.

We pursue three questions: (i) What statistical and correlational regimes govern engineered features? (ii) How do individual physics drivers map onto an inferred continuous habitability score? (iii) Which features emerge as most influential in a tree-ensemble interpretability context?

2 Data and Feature Engineering

We aggregate cleaned mission and archive sources (NASA Exoplanet Archive, TESS TOI supplements, Gaia cross-matches) and compute derived quantities:

- **Equilibrium Temperature:** $T_{eq} = T_\star \sqrt{\frac{R_\star}{2a}}(1-A)^{1/4}$ under zero greenhouse and uniform reradiation assumptions.
- **Insolation Flux:** $S = \dfrac{L_\star}{4\pi a^2}$, scaled to Earth units to contextualize climatic forcing.
- **Escape Velocity Ratio:** $v_{esc}/v_{esc,\oplus} = \sqrt{(M/M_\oplus)/(R/R_\oplus)}$, a proxy for atmospheric retention probability.
- **Tidal Locking Proxy:** Timescale τ_{lock} approximated (order-of-magnitude) by $\tau \propto \dfrac{a^6 Q}{3Gm_\star^2 k_2 R^5}$ (constants suppressed for relative comparison).
- **Earth Similarity Index (ESI):** Composite similarity metric over normalized radius, density, and temperature dimensions.

3 Methods

We perform a staged visual-analytic protocol: (a) univariate distribution assessment (Figure 1), (b) correlation analysis (Figure 2) via Pearson coefficients r_{XY}, (c) multi-driver response grids (Figure 3), (d) binned trend smoothing (Figure 4), (e) margin evaluation between discretized classes (Figure 5), and (f) ensemble feature importance (Figure 10). Random Forest importance leverages impurity reduction: Gini $I_G = \sum_k p_k(1-p_k)$; cumulative weighted decreases approximate marginal contribution.

4 Results

4.1 Distributional Structure

Figure 1 reveals scale heterogeneity and skewness: long tails in irradiation and temperature derivatives contrast with relatively compact distributions of size ratios. Such disparities motivate normalization and caution against variance-dominated learning.

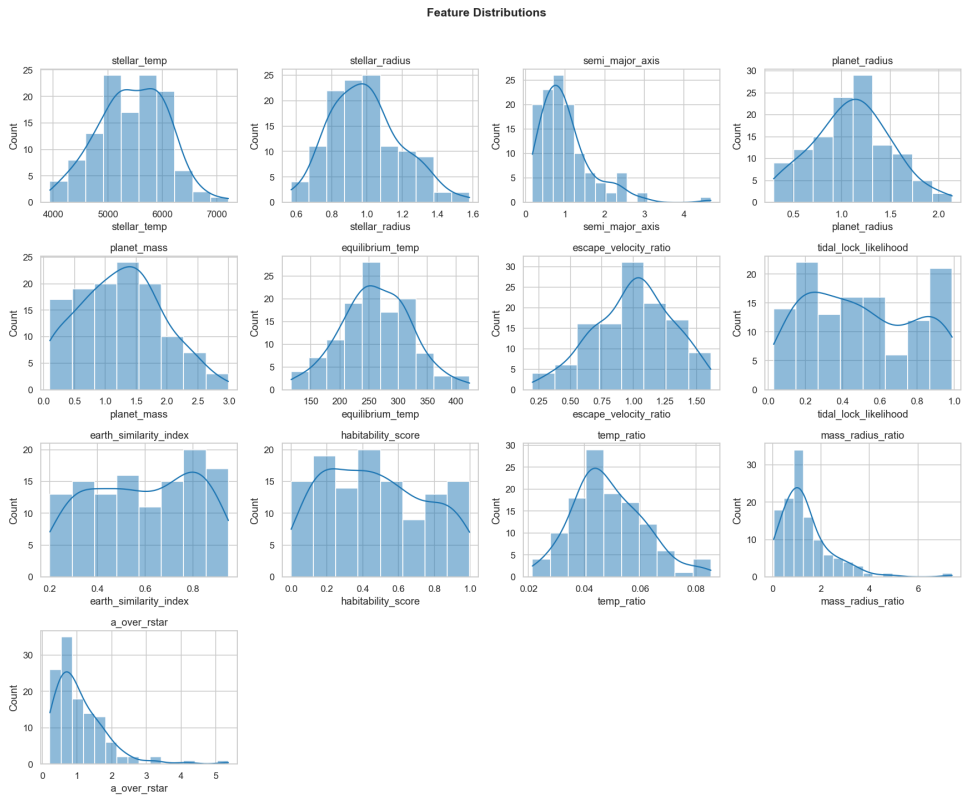

Figure 1: Univariate feature distributions highlighting skew, multi-modality, and scale disparities relevant to model calibration.

4.2 Correlation Regimes

Figure 2 indicates clusters (e.g., radius, mass-derived gravity proxies) exhibiting high linear dependence, implying potential dimensionality reduction. Moderate correlation between flux and equilibrium temperature underscores partially redundant radiative descriptors.

4.3 Driver Response Surfaces

The multi-panel grid (Figure 3) illustrates monotonic or threshold-like relationships between physics drivers and habitability score. Saturation suggests diminishing returns for certain radiative metrics beyond optimal mid-range bands.

4.4 Binned Trend Aggregation

Figure 4 smooths noise via bin means $\bar{y}_j = \frac{1}{|B_j|} \sum_{i \in B_j} y_i$, clarifying coarse functional shapes (e.g., peaked vs strictly increasing), aiding feature transformation considerations.

4.5 Score vs Class Margin

The separation visualization (Figure 5) assesses overlap between class-conditioned score densities, constraining achievable classification performance and guiding threshold calibration.

4.6 Fundamental Scaling: Temperature vs Distance

Figure 6 tests radiative equilibrium scaling $T_{eq} \propto a^{-1/2}$. Deviations may arise from albedo assumptions and stellar parameter uncertainties.

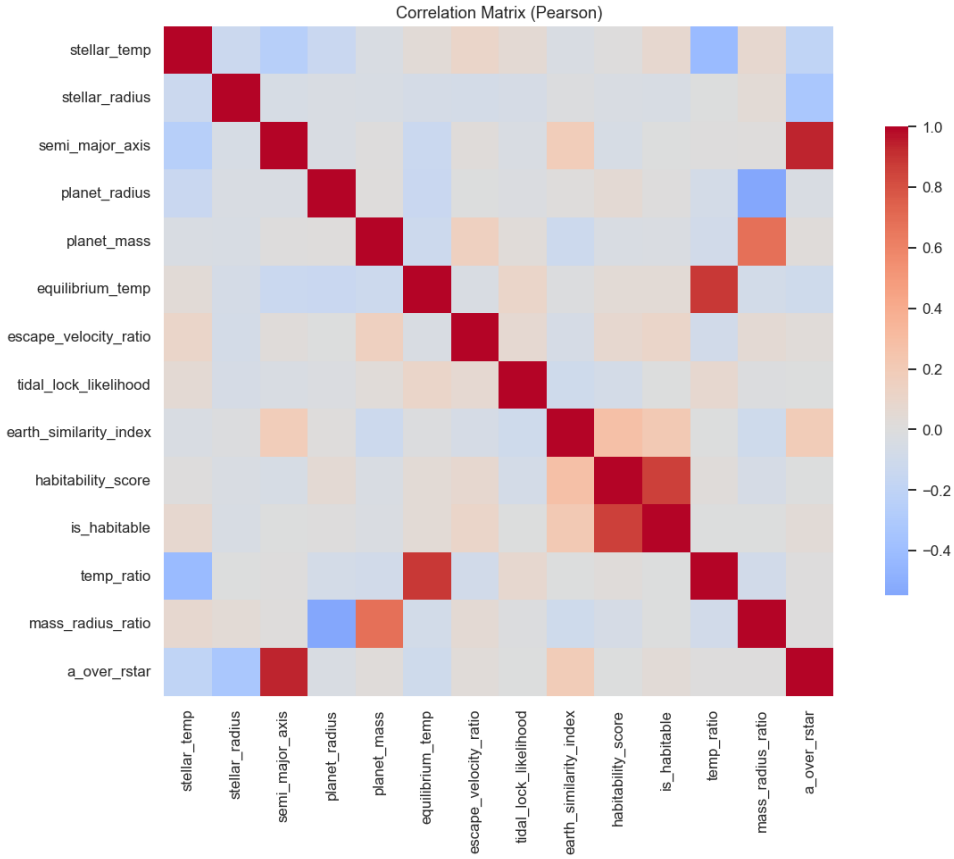

Figure 2: Pearson correlation matrix showing redundancy blocks and informing feature pruning strategies.

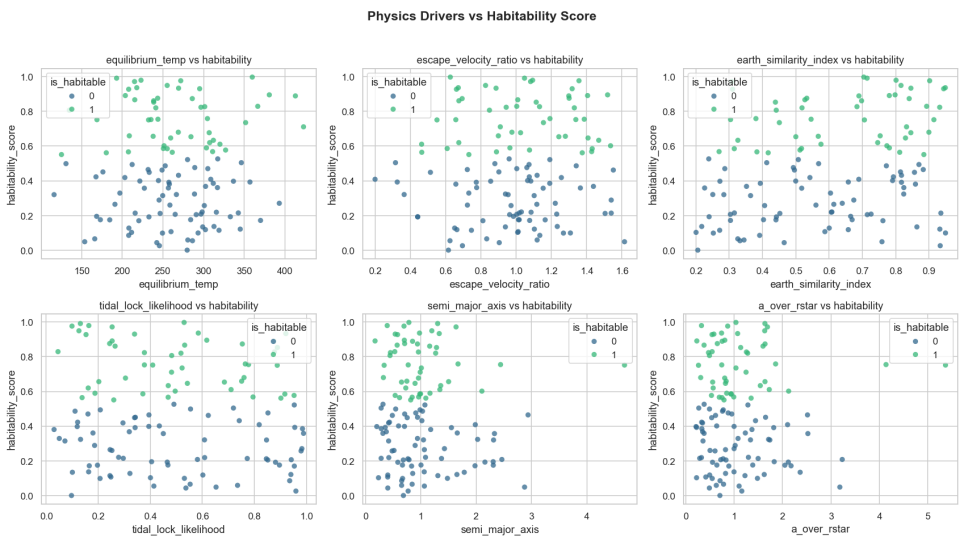

Figure 3: Scatter grid mapping individual physics-derived features to inferred habitability score, revealing monotonic and saturating behaviors.

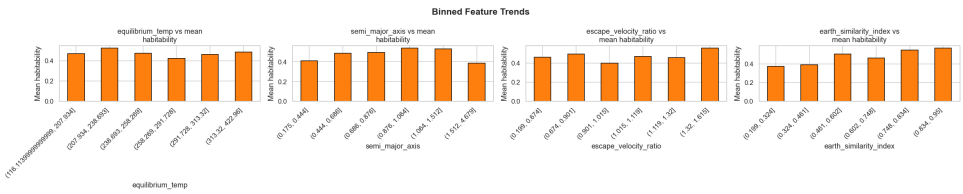

Figure 4: Binned averages emphasize macro-level tendencies while attenuating local stochastic variance.

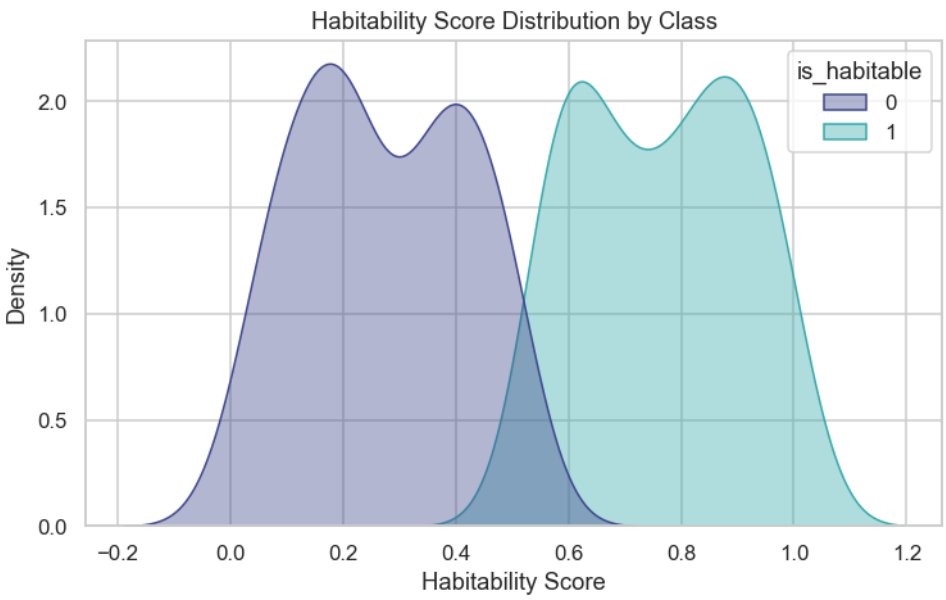

Figure 5: Continuous habitability score distributions stratified by class label; overlap regions bound discriminative ceiling.

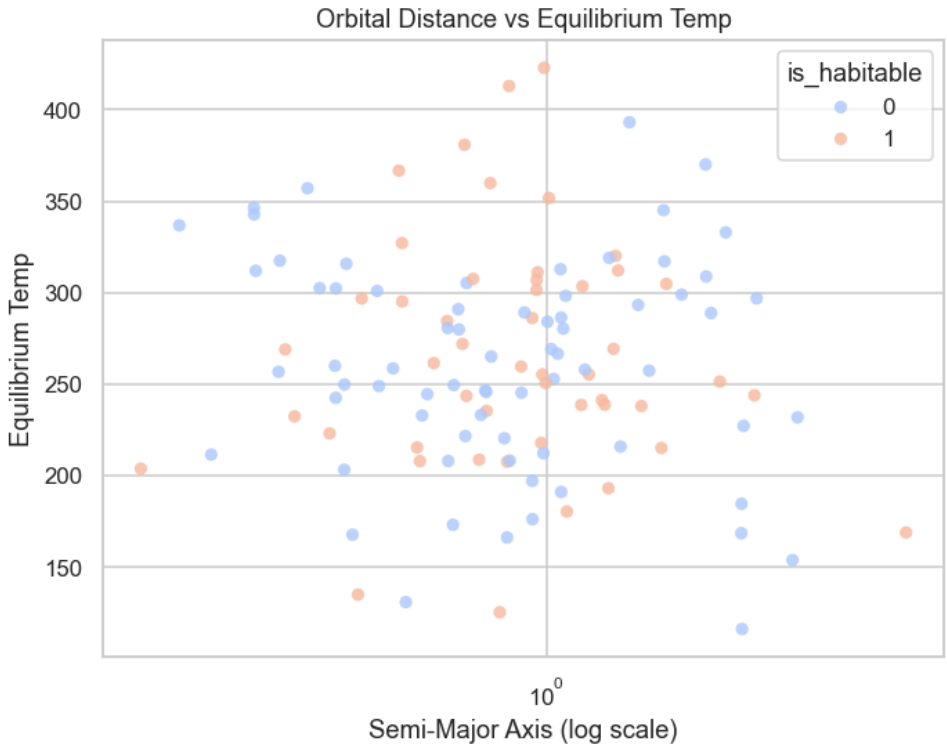

Figure 6: Equilibrium temperature trend versus semi-major axis showing inverse square-root qualitative behavior with residual dispersion.

4.7 Escape Velocity and Retention Potential

Figure 7 shows a mild positive association: higher v_{esc} correlates with improved retention probability, mitigating volatile loss and enabling stable atmospheric envelopes.

4.8 Composite Similarity (ESI)

Figure 8 validates that higher ESI distributions skew toward elevated habitability scores, affirming coherence between composite similarity logic and learned outcomes.

4.9 Tidal Lock Likelihood

Figure 9 evaluates rotational state implications. Elevated locking likelihoods may introduce longitudinal climatic extremes, reflected in modestly reduced central habitability tendencies.

4.10 Model-Derived Feature Importance

Figure 10 ranks features via Random Forest impurity reduction, emphasizing radiative and retention proxies; correlated clusters diffuse individual attribution.

5 Discussion

Key findings: (i) Radiative balance variables dominate explanatory power; (ii) Redundant bulk descriptors invite dimensionality compression; (iii) Atmospheric retention proxies contribute secondary, yet consistent, uplift; (iv) Tidal locking indicators show nuanced, non-binary influence requiring more granular dynamical modeling. Limitations include simplified albedo assumptions and neglect of stellar activity modulation.

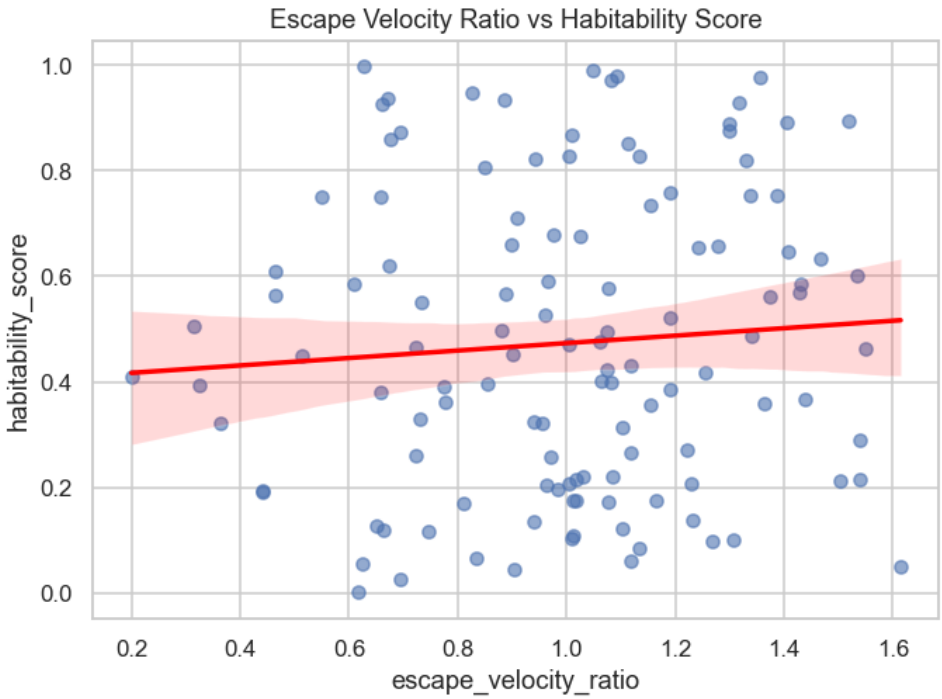

Figure 7: Relationship between escape velocity and habitability score, reflecting atmospheric retention influence.

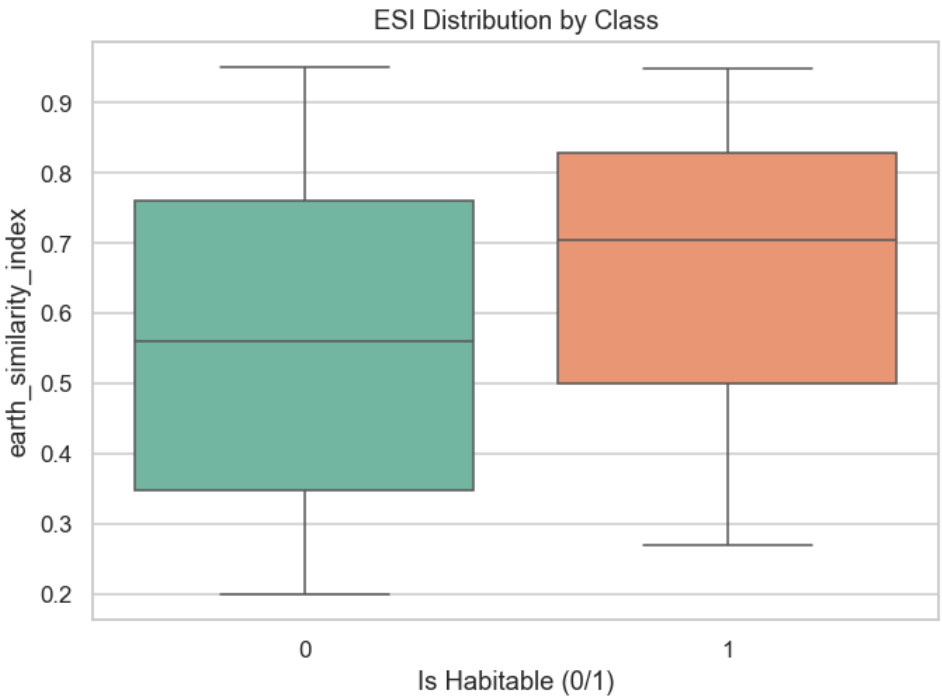

Figure 8: Earth Similarity Index distribution across habitability class partitions.

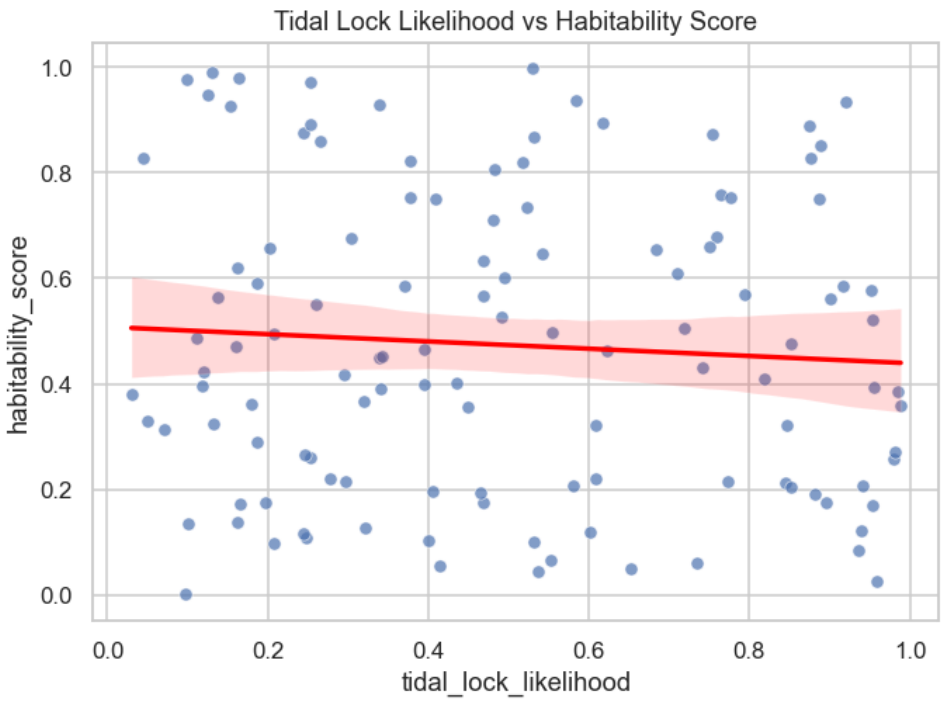

Figure 9: Habitability score variation with a tidal locking proxy; potential climatic dichotomy effects noted.

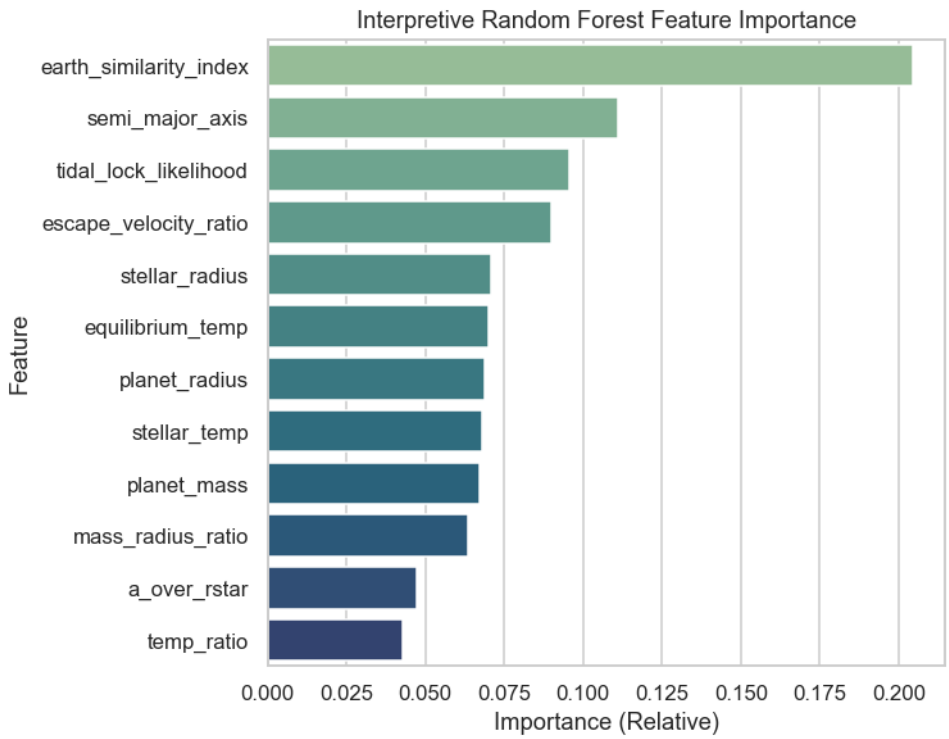

Figure 10: Relative feature importance indicating dominant physics-derived explanatory variables.

6 Limitations and Future Work

Future enhancement pathways: incorporation of stellar flare frequency statistics, photochemical atmospheric loss modeling, spectral biosignature priors, and Bayesian hierarchical approaches for propagating measurement uncertainties into habitability inference.

7 Conclusion

This refined academic treatment links first-principles physics to predictive habitability modeling through a structured visual-explanatory lens, providing an interpretable scaffold for subsequent spectroscopic and temporal expansion.

Acknowledgements

We acknowledge open data assets provided by NASA archives and community-maintained exoplanet catalogs.

References